Chemistry 3A

Introductory General Chemistry

Experiment 1a

Making Measurements

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Introduction

Measurements of mass and of length in one dimension and length in three dimensions to get volume are common in the laboratory

In this experiment you determine mass and volume and length measurements on a metal bar and on pure aluminum shot to get the property of density

Is the metal bar aluminum or another metal based on your calculations?

Background

Measuring mass. We did this last week with small amounts of table salt (sodium chloride, NaCl) and of DI water using the electronic balance. An electronic balance has a maximum capacity. Other scales, like a (triple-)beam balance are used for heavier objects



Measuring volume. We need an accurate volume measurement. It will be done in two ways:

1. Using a ruler to measure the lengths and calculate usually in three dimensions (width, height, breadth)
2. Displacement of water in a measuring container such as a graduated cylinder. Sufficient water to fully immerse the object is filled in the cylinder and an initial reading taken. The object is immersed in the cylinder with water (it MUST be completely immersed), then the final reading is taken.

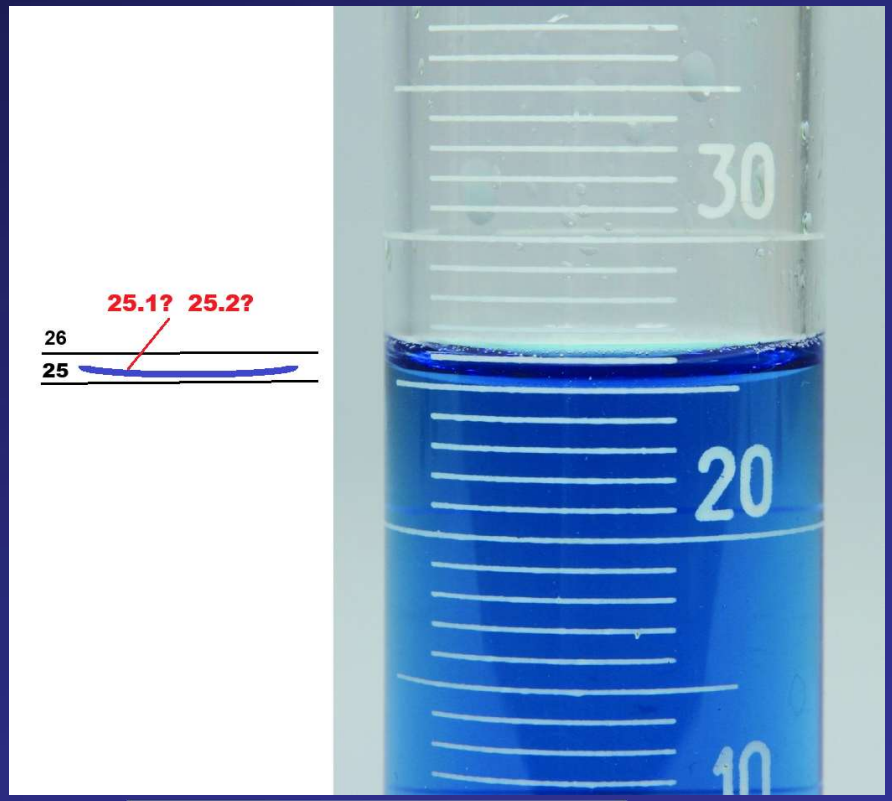
A Note About Reading Precision

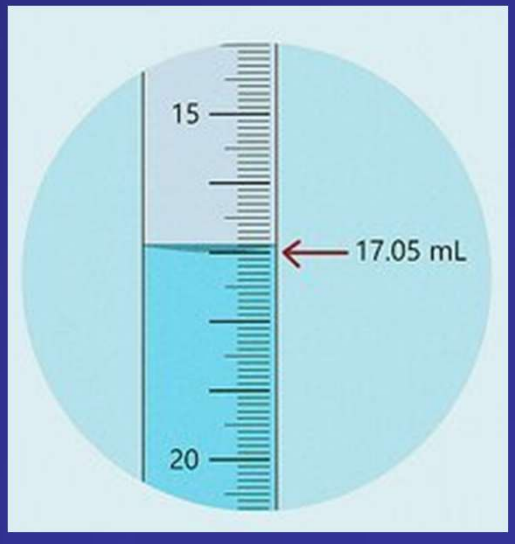
Volumetric devices / instruments / containers like a buret and graduated cylinder have markings (the major markings have numbers)

It is asserted that if an instrument has 1 mL markings like a graduated cylinder, it is precise to one-tenth of that, that is, ±0.1 mL

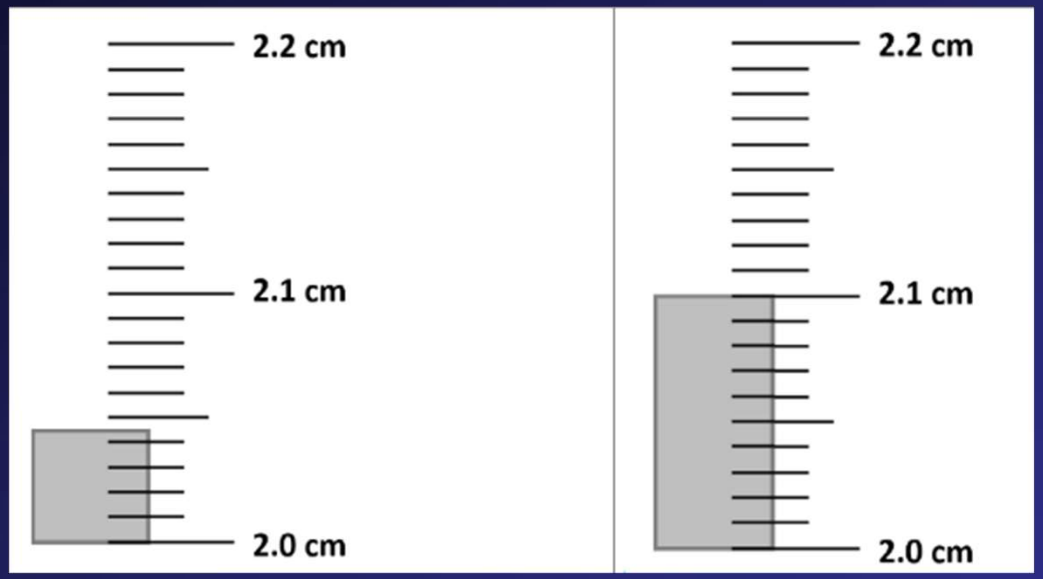
If it has 0.1 mL markings like a buret, it is precise to one-tenth, or ±0.01 mL

User your best judgment





Notes on Measurement Precision



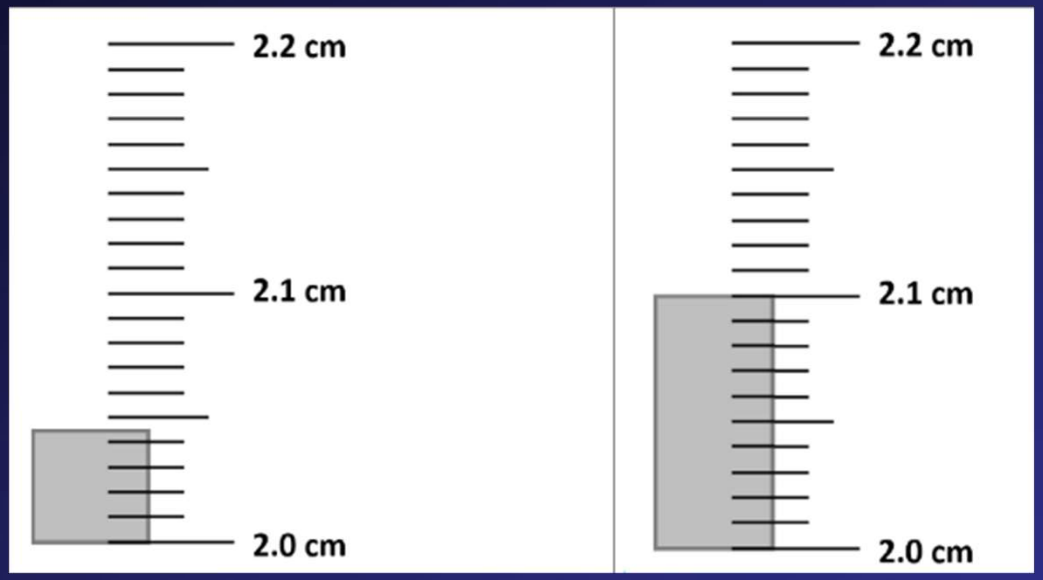
- Look at the smallest division mark. What is its value?

  1 mL? 0.1 mL? 0.01 mL? 0.001 mL?

- Now take one-tenth of the division mark value: multiply by 0.1 or by 10%. Or divide by 10

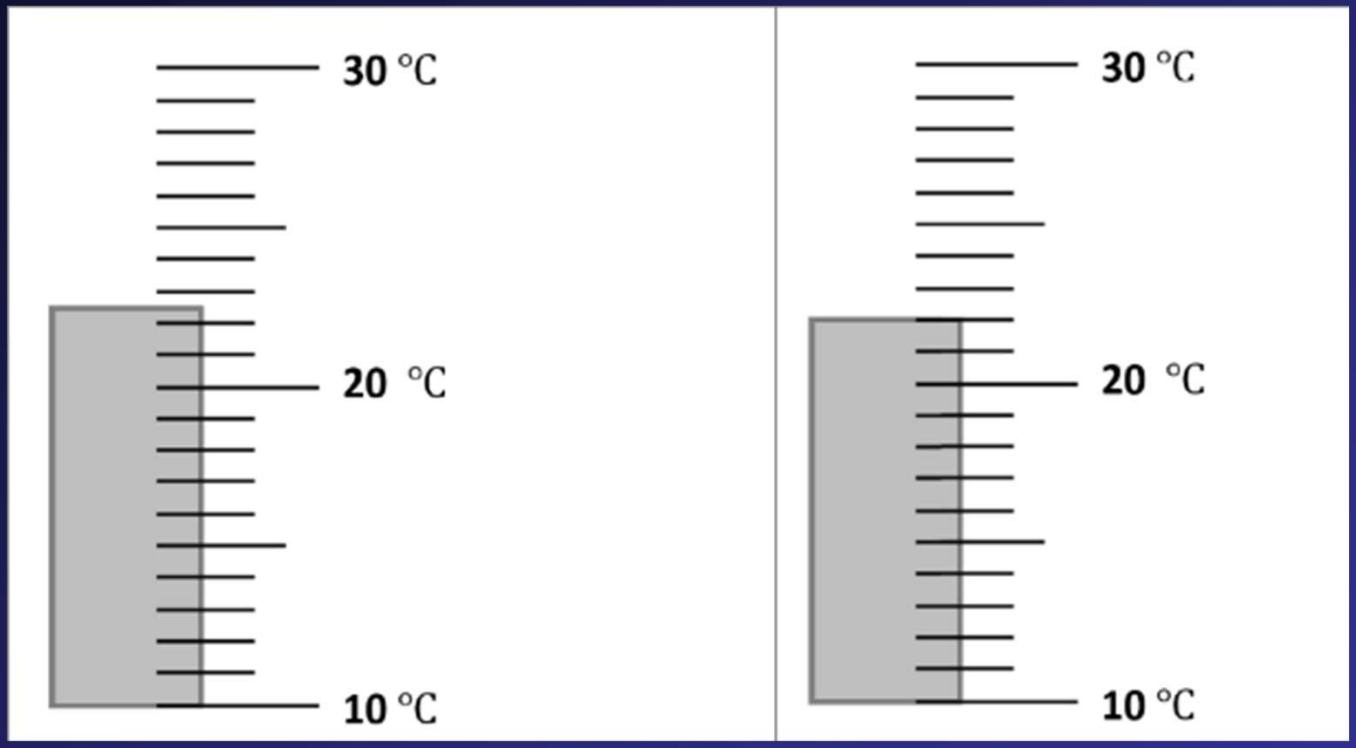
  That is your precision!

Notes on Measurement Precision



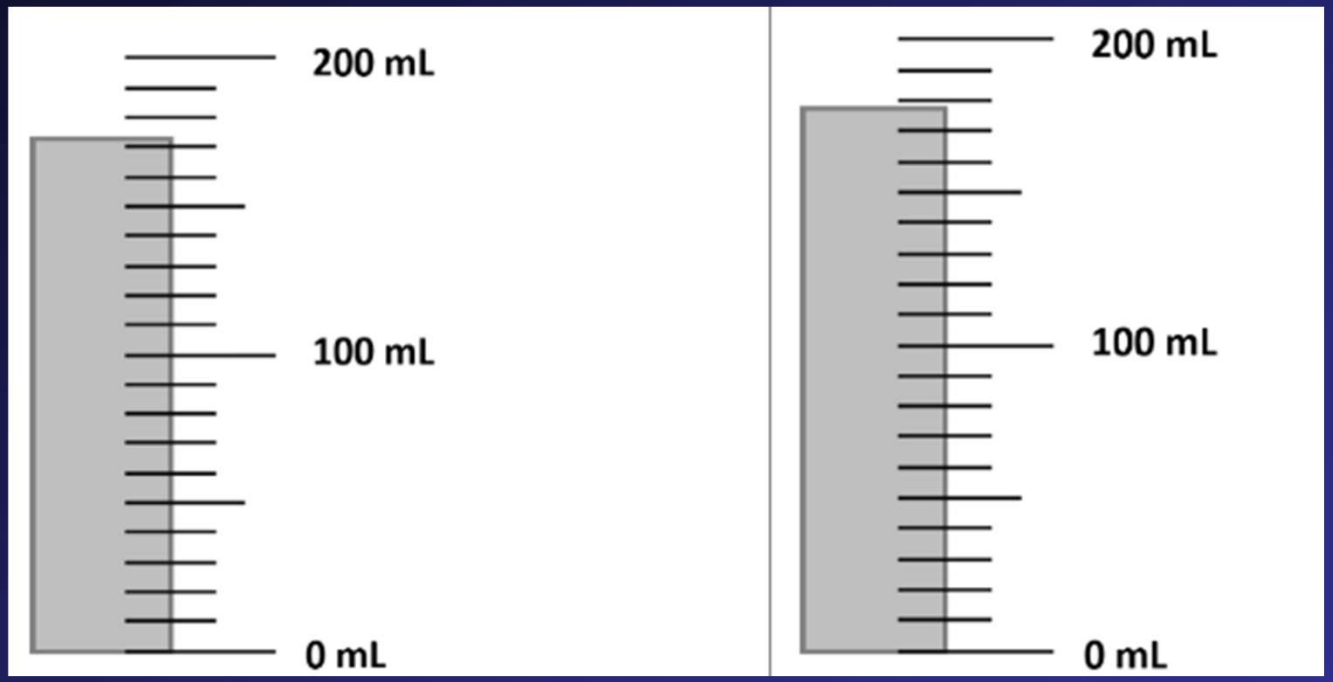
- *left*
  Division mark: 0.01 cm
  Precision = 0.01 cm x 10% = 0.001 cm
  Measure = **2.045 cm** ← precise to 0.001 cm!
- *right*
  Division mark: same as 1(a) → 0.01 cm
  Precision = same as 1(a) → 0.01 cm x 10% = 0.001 cm
  Measure = 2.10 cm but precise to 0.001 cm → **2.100 cm**

Notes on Measurement Precision



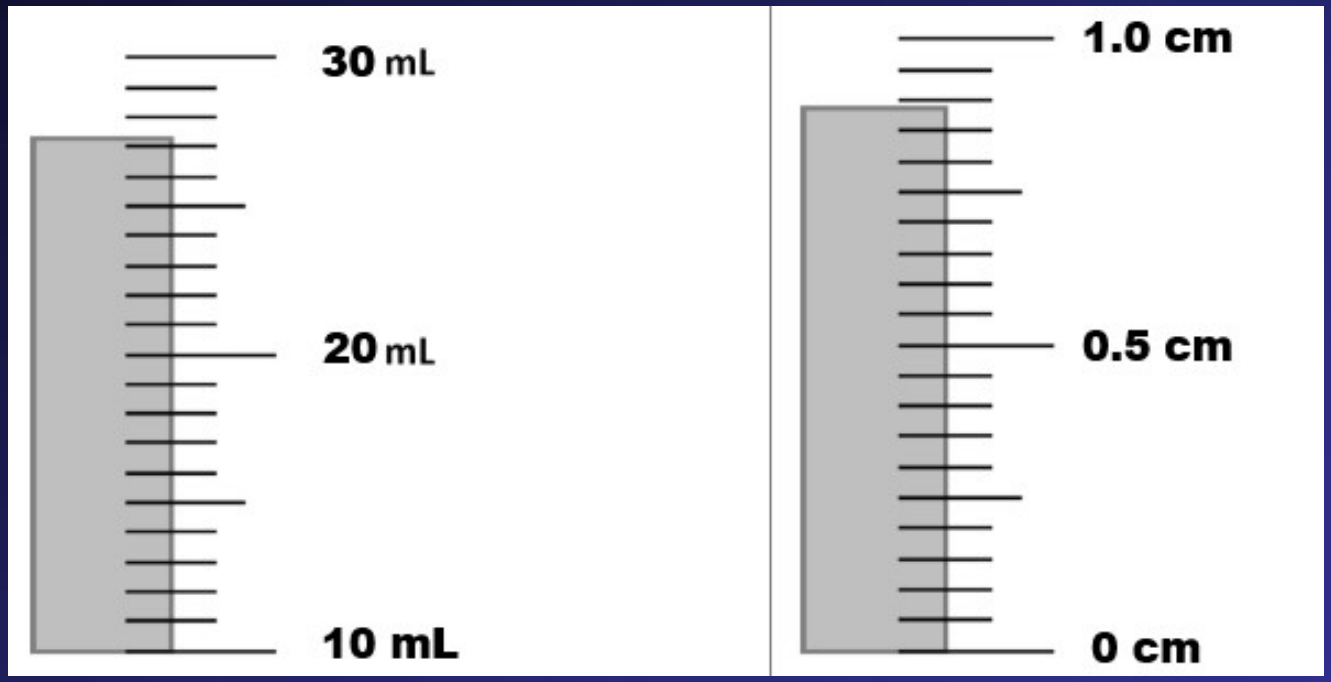
	Division mark	Precision	Measure	
left	1°C	0.1°C	22.5°C	Precision = division mark x 10%
right	1°C	0.1°C	22.0°	Not "22°C"

Notes on Measurement Precision



	Division mark	Precision	Measure	
left	10 mL	1 mL	172 mL	Not "170 mL" or "172.0 mL"
right	10 mL	1 mL	178 mL	

Notes on Measurement Precision



	Division mark	Precision	Measure	
left	1 mL	0.1 mL	27.2 mL	
right	0.05 cm	0.005 cm	0.880 cm	precise to 3rd decimal digit

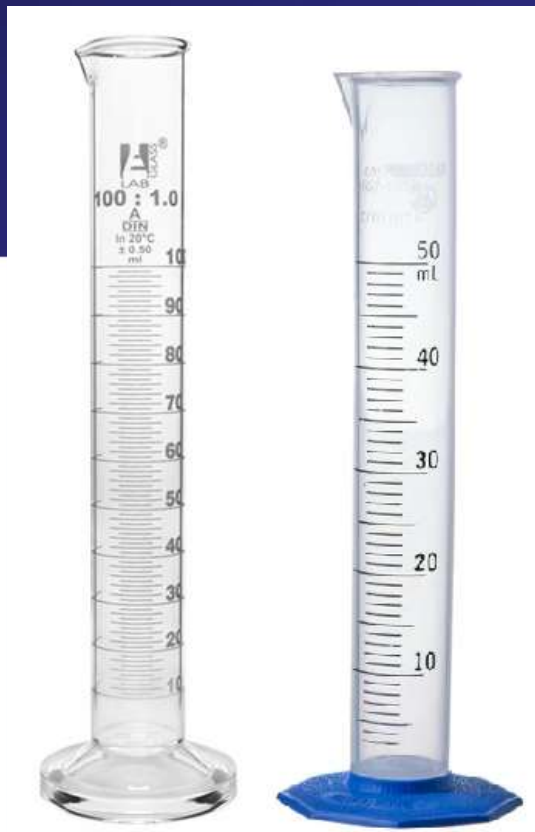
Equipment You Will Use



electronic balance

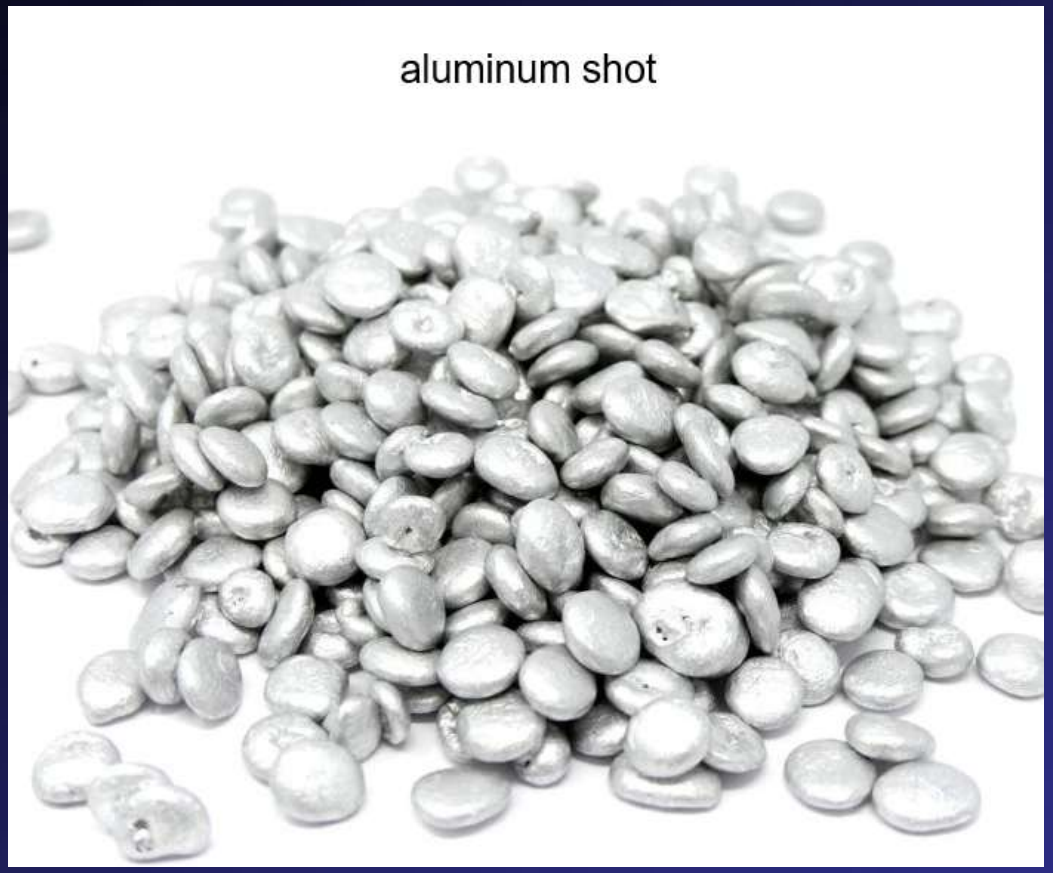


glass stirring rod



Equipment You Will Use

aluminum shot

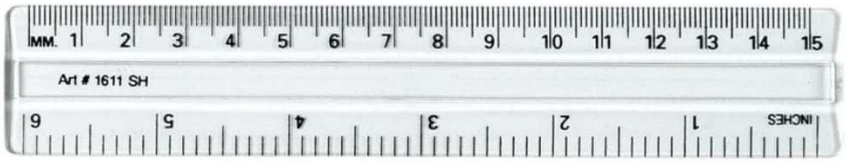




metal bar



metric ruler



Procedure

Equipment/Material Needed

weighing balance	metric ruler
50 or 100 mL graduated cylinder	10-30 g aluminum shot
glass rod	

Consumables Needed

DI water (~60 mL)

- p. 22: record identifier on metal bar; describe its appearance
- measure bar mass. Record all digits on balance
- Use ruler to measure bar dimensions; use significant digits
- Use water-filled graduated cylinder to measure volume. Make sure it is fully immersed! Make sure to use glass rod to tap out air bubbles on any object measured by volume displacement method
- Record your data and make your calculations, taking care to express results using significant digits

Clean Up

- Use paper towel to dry the aluminum shot, then put in the “return metal” container. Also dry the metal bar before putting in container
- Empty the water bottle before putting it back
- Place all other equipment from where it was obtained

Example Data Analysis

DATA

1. Number or letter on the bar: 37
2. Mass and appearance of the metal bar.
   a. Mass 72.1773 g
   b. Shape rectangular prism
   c. Color silver
3. Dimensions of the bar:
   a. Longest side (Length) – or Diameter (for cylinders): 3.65 cm
   b. Medium side (Height): 1.70 cm
   c. Short side (Width): 1.05 cm (not needed for cylinders)
4. Water displacement of the bar:
   a. Initial Water Volume 26.0 mL
   b. Final Water Volume 35.0 mL
5. Mass and appearance of the aluminum shot:
   a. Mass 14.2880 g
   b. Shape pellets
   c. Color silver
6. Water displacement of the aluminum shot:
   a. Initial Water Volume 20.0 mL
   b. Final Water Volume 26.0 mL

precision:
- for cm measurements typically two decimal places! (0.00 cm)
- for masses: record all numbers on balance display!
- for 50 or 100 mL graduated cylinders: one decimal place (0.0 mL)

if object is cylinder,
use 3a for diameter and
3b for height (only two measures)

- "mL" NOT "ml"
- volume calculations:
  -- for solid matter/objects: "cm^3"
  -- for liquids: "mL"
- conversion factor: "1 mL/1 cm^3" or "1 cm^3/1 mL"
- object with 3 dimensions:
  V = length (cm) × width (cm) × depth (cm) = object volume (cm^3)
- cylindrical object:
  V = pi × [square of radius (cm)] × height (cm) = $\pi r^2 h$ = cylinder volume (cm^3)
- diameter = twice radius = $d = 2r$
- radius = half diameter = $r = d/2$

Example Data Analysis

Exp 1a (Intro) *Name:*

RESULTS

* Show your work and keep track of significant figures.

- quantities are numerical values that 99.85% of the time also have units
- quantities are measurements and calculations

7. Calculate the metal bar's volume (in cm^3) using its dimensions.

$$3.6\ cm \times 1.8\ cm \times 1.2\ cm = 7.8\ cm^3$$

- This lab book has questions or items with a lot of whitespace
- This whitespace is the strong hint that you should NOT only record an answer/result, but that you should SHOW YOUR WORK, SHOW YOUR REASONING!
- Includes writing math expressions

8. Calculate the volume of the bar in mL using water displacement.

$$35.0 - 26.0 = 9.0\ mL$$

9. Chemists compare two values mathematically by calculating percent difference. Calculate the percent difference between the two volumes using the following equation:

$$\%\ Difference = \frac{Volume\ from\ dimensions - Volume\ by\ displacement}{Volume\ by\ displacement} \times 100$$

- This is a **conversion factor**
- These are used heavily in chem math

$$\frac{7.8\ cm^3 \times \frac{1\ mL}{1\ cm^3} - 9.0\ mL}{9.0\ mL} \times 100\% = -13.3\%$$

- Conversion factors should be presented to cancel one unit to create another

10. How does the bar's 'volume measured by dimensions' compare to the bar's 'volume measured by displacement'? (Use complete sentences, mentioning both numerical volume measurements, and the percent difference)

bar volume as measured by ruler (7.8 mL) is smaller than measured by water displacement (9.0 mL), the ruler measured 13.3% less

Example Data Analysis

Exp 1a (Intro) *Name:*

11. Calculate the density of the metal bar. (Use the volume from the displacement measurement.)

$$\rho = \frac{\text{mass}}{\text{volume}} = \frac{72.1773\ g}{9.0\ mL} = 8.0\ g/mL$$

Math operations on quantities
- you can do multiplication and division operations on quantities with different units
- you can not add or subtract quantities with different units; they must be the same or you have to convert so units are same
- calculated results of multiplication/division operations will have units f all quantities in numerator & denominator as they were evaluated
- density the computation of quotient: mass / volume, so it has units of mass divided by units of volume, such as "g / mL" or "kg / L"

12. Determine the density of the aluminum shot.

$$\rho = \frac{14.2880\ g}{6.0\ mL} = 2.4\ g/mL$$

13. Compare the densities of the aluminum shot to the density of the metal bar you measured earlier. Could your bar be made of aluminum? Why or why not? If the bar is not aluminum, what could the metal be?

Metal	Density (g/cm^3)
Brass	8.5
Cast Iron	6.7
Copper	9.0
Gold	19.3
Lead	11.3
Silver	10.5
Stainless Steel	7.5
Steel	8.0
Titanium	4.5
Tin	5.8
Aluminum	2.4

difference in density between unknown metal and aluminum shot is about 3-fold and too large for unknown to be aluminum

density of unknown is an exact match for "steel" in the table, so this is most likely the identity of the unknown metal

This experiment effectively added your determined density value of **aluminum** to the table for your use